# DASC 3240, L12

## More plots – Assignments

2023/24 SPRING SEMESTER

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# Data

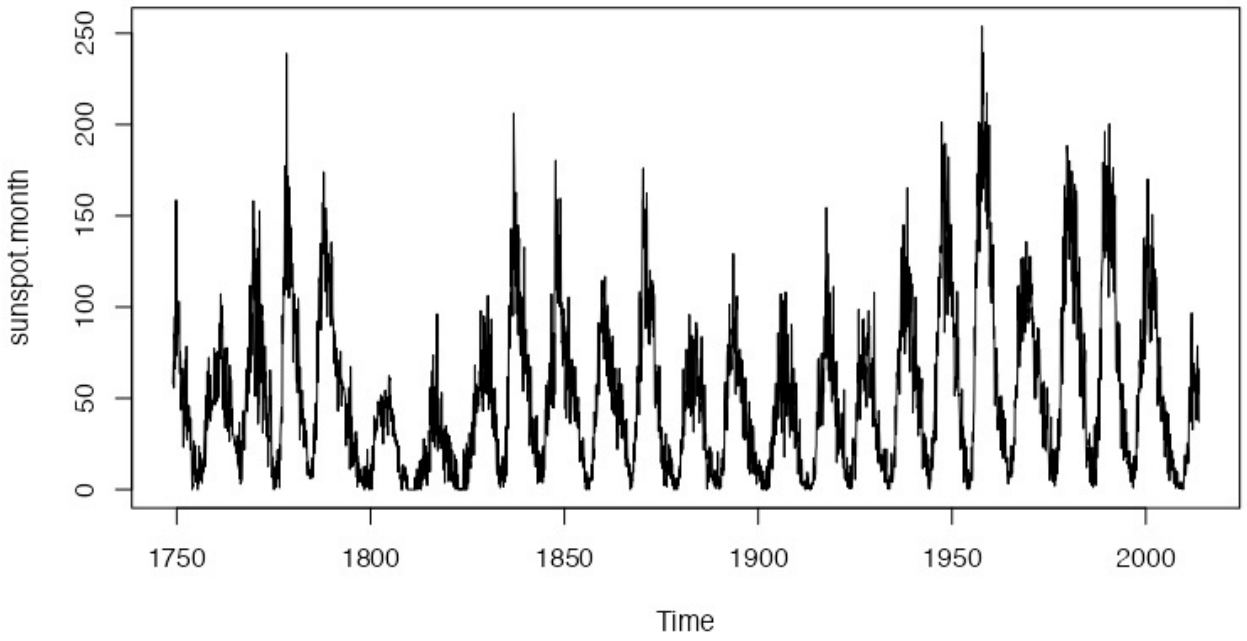
# sunspot.month

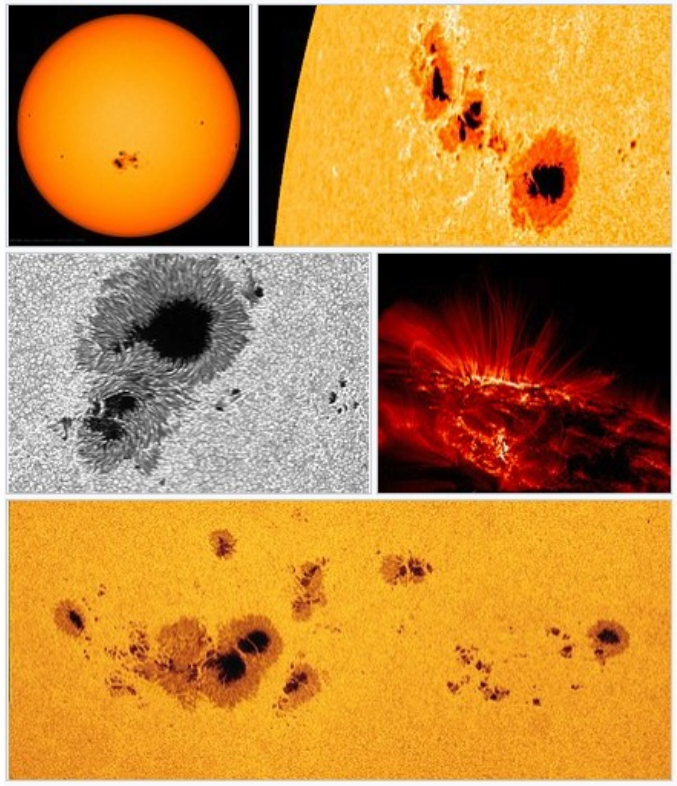
https://vincentarelbundock.github.io/Rdatasets/doc/datasets/sunspot.month.html

## Monthly Sunspot Data, from 1749 to "Present"

### Description

Monthly numbers of sunspots, as from the World Data Center, aka SIDC. This is the version of the data that will occasionally be updated when new counts become available.



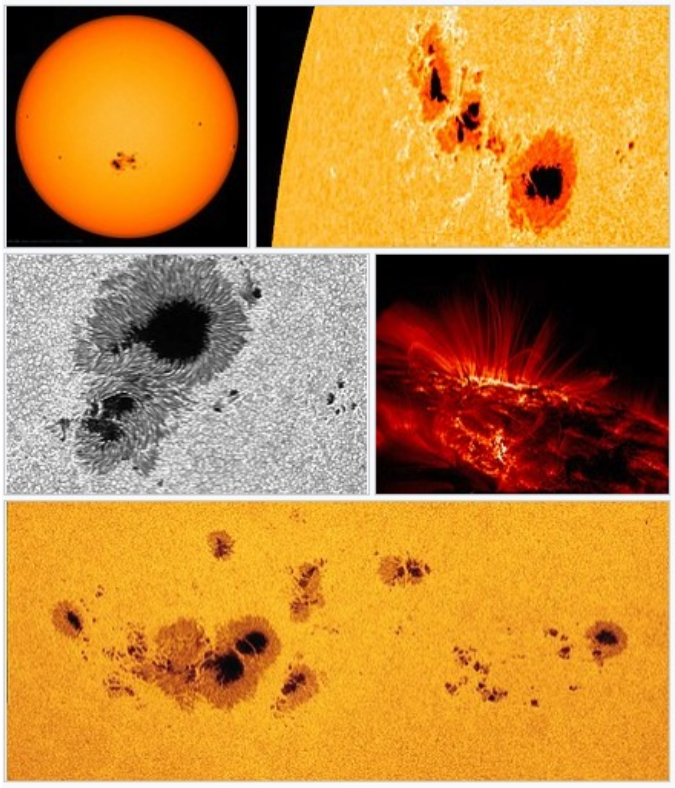
https://en.wikipedia.org/wiki/Sunspot

# sunspot.month

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/sunspot.month.html

**I modified the original data slighly and put the modified data file in the DASC 3240 GitHub repository.** Please download the file and use it for the assigment.

https://github.com/ong8181/DASC3240/blob/main/data/sunspot_month_L12.csv

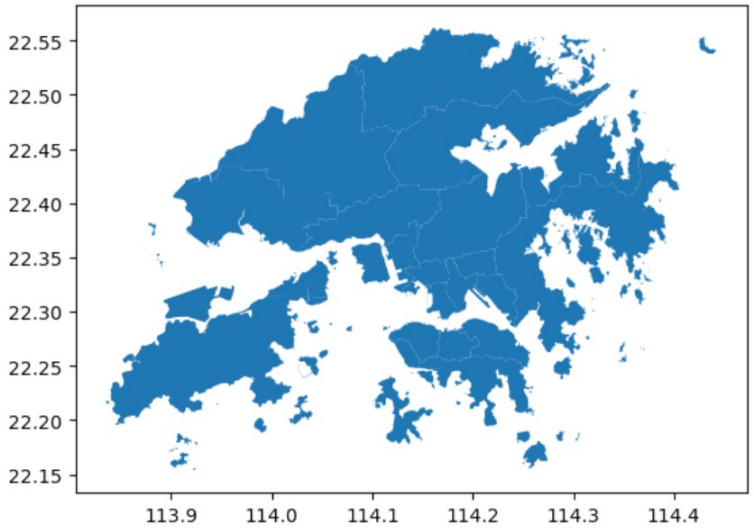


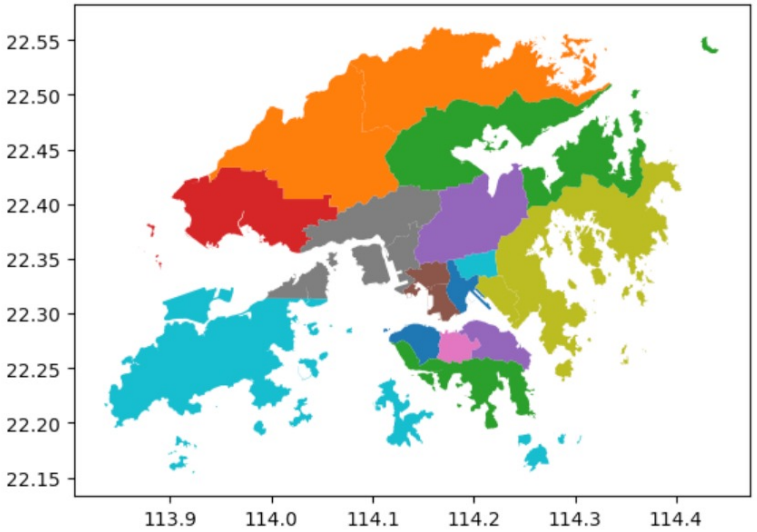
https://en.wikipedia.org/wiki/Sunspot

# Map

Hong Kong map data is available from the following websites.

- DATA GOV HK | https://data.gov.hk/en-datasets
- Open Geospatial Data in HK | https://opendata.esrichina.hk/



# Assignments

# Assignments

- **sunspot_month_L12.csv**
  1. Load the data and visualize the time series of the monthly sunspot patterns.
  2. It is known that the number of sunspots fluctuates in an 11-year cycle. Visualize this by drawing vertical lines to the graph every 11 years.
- **Map**
  3. Check Hong Kong .geojson (or .json) files listed below. Draw a HK map using the geopandas package and the file.
     - Hong Kong geojson file | https://opendata.esrichina.hk/maps/hong-kong-18-districts
     - Other source | https://data.gov.hk/en-datasets
     - [I will **NOT** mark this] Also, please check the license of the file (For what purpose can we use it?).

[**Optional**] If you can find free HK `.geojson` or `.json` files, describe URLs with their license (e.g., CC BY).